

Plaksha University
DS3005
Computational Algorithms in Data Science
Project Guidelines

1 Project proposal

1. The project proposal carries 5% weightage in the overall evaluation. The proposal should include the following components:
 - Motivation and background of the topic
 - A brief review of the latest (preferably after 2015) literature/papers (at least 2) in which your topic is implemented/modified/improved/ altered/generalized.
 - Precise research statement: What technique/algorithms do you intend to study? How will you demonstrate the applications of the methodology?
 - Expected results and learning outcomes: What outcomes are expected? How will those be useful in which context? How does the project topic align with the classroom learning in our course?
 - Technical details: Mathematical and methodological details of the proposed topic: The algorithms and formulas, assumptions, computational toolkit (software and functions) for implementation (No need to show any codes or results, tables, graphs in the proposal).
 - Data availability statement: Whether the data are publicly available in repositories (share the link)? Whether you will request the authors of papers you have read to make their data available? Whether you will use the synthetic data simulated from the models in any of the papers you have read? Whether you plan to collect the primary data through surveys?
 - AI-usage and plagiarism statement: What AI tools do you plan to use for what purposes? How will you ensure the originality of the work at the backdrop of AI-usage? How will you prevent producing duplicated/plagiarized results?
2. The proposal of each team will be reviewed and comments will be given by me and an additional expert. Each team should implement the suggested changes and corrections.
3. The revised proposals should be sent back along with a self-contained document/report mentioning concisely how you responded to each comment and what steps you took to address the suggested changes. This report is important since it gives an overview of your thought process for addressing the comments on your proposal.

4. Typically, if you diligently make the corrections and supplement the revised proposal with a report of your responses on each comment, your proposal will be approved and you can go ahead with the actual work.
5. In exceptional cases, the proposals may be subjected to additional revisions if earlier revisions are insufficient, inadequate or incomplete.
6. You can start working towards the final report only after your proposal has been approved.
7. If your proposal needs multiple revisions, the beginning of your actual work will get delayed and you may fall behind the other teams. Hence, please ensure that you make the proposal sincerely and attempt to address all the comments thoroughly in the very first go.

2 Project report

1. The project report needs to be submitted in two stages: Stage 1-Interim report and Stage-2: Final report
2. Both the versions should have the following components:
 - Introduction, background and motivation
 - Problem statement
 - Technical details of the algorithm including methodology, formulas and assumptions
 - Data description and source
 - Results of the analysis
 - Conclusions and key observations
 - Overall learning outcomes
 - Limitations of the proposed study
 - Future directions and scope for further investigations
 - Appendix A: Codes
 - Appendix B: History of AI prompts and chats.
 - Appendix C: A document showing one-one correspondence of the AI-chats and prompts with the codes and outputs in the report. This should clearly highlight which part of the AI-generated output is used where and for what in your report.
 - Appendix D: Additional technical details, graphs, images and supplementary data (if any). This appendix is optional.
3. The interim report should be submitted within 2 weeks after the approval of your proposal.

4. The interim report carries 5% weightage.
5. The interim report will be reviewed and comments will be given. These should be implemented in the final report.
6. The final report should be submitted within 2 weeks after receiving the comments on the interim report.
7. The final report should be accompanied by the report on how you addressed the comments on the interim report. This document is similar to the report that you are supposed to send along with the revised proposal.
8. The final report is NOT simply the corrected and refined version of the interim report. It should also have additional modeling results and output not covered in your interim report. As such, the final report is a superset of the interim report.
9. The final report carries 10% weightage.

3 Presentation and open viva

1. The project should be presented during the end-semester examination.
2. All the members of the team should present for more or less same duration.
3. The presentation will be followed by a viva and Q&A. The examiners will be me and an additional external expert.
4. After addressing the questions and comments by the external examiner, the floor will be open to selective questions from the audience.
5. All the team members are required to participate equally in the viva and interaction.
6. The weightage for the presentation and viva component is 10%.
7. The scores of this component may be different for different team members depending upon the nature, quality and accuracy of their interaction and responses to the viva questions.

4 Component-wise weightage

The project component carries 30% weightage in the overall assessment of the CADS course. The weightage of the each sub-component is follows.

Component	Weightage
Proposal	5%
Interim report	5%
Final report	10%
Presentation and viva	10%
Total	30%

5 Important dates and deadlines

Task	Date/deadline
Proposal submission	Due on 27th March 2026 by 6PM
Comments on the proposal	Will be sent by 3rd April 2026 by 6PM
Revised proposal submission with a report on addressing the comments	Due on 7th April 2026 by 6PM
Interim report submission	Due on 21st April 2026 by 6PM
Comments on the interim report	Will be sent by 24th April 2026 by 6PM
Final report submission	Due on 7th May 2026 by 6PM
Presentation and open viva	During the end semester examination on the day(s) allotted by OAA